Sanskrit Translation Datasets

अनुवाद डेटासेट संकलन

Comprehensive Guide to Free Sanskrit-English Parallel Corpora & Dictionaries

Curated for Neural Machine Translation (NMT), LLM Fine-Tuning, and Lexical Alignment

Sanskrit-to-English translation presents unique computational challenges, including rich inflectional morphology, extensive Sandhi compounding, and flexible free word order. This catalog gathers the most authoritative, free, and publicly accessible parallel corpora, bilingual lexicons, grammatical databases, and pre-trained translation models to train, fine-tune, or benchmark high-accuracy Sanskrit machine translation pipelines.

1. Large-Scale Sanskrit-English Parallel Corpora (Machine Translation)

Pre-aligned sentence pairs for training and evaluating Transformer and Seq2Seq NMT models

1.1 buddhist-nlp/skt-en-itihasa (Hugging Face)

Domain: Mahabharata & Ramayana Classical Epics | Volume: 93,000+ parallel sentence pairs (Train: 89.4k, Val: 1.8k, Test: 1.8k)

Link: https://huggingface.co/datasets/buddhist-nlp/skt-en-itihasa

The largest cleanly aligned Sanskrit-English classical literary corpus available on Hugging Face. Covers complete narratives with human-verified translations and Devanagari punctuation.

Usage: Python: from datasets import load_dataset; ds = load_dataset('buddhist-nlp/skt-en-itihasa')

1.2 AI4Bharat IndicTrans2 Sanskrit Parallel Corpus

Domain: Mixed Domain (Conversational, Literary, Formal) | Volume: 45,000+ high-quality parallel pairs (san_Deva <-> eng_Latn)

Link: https://huggingface.co/datasets/ai4bharat/indic-trans-v2-raw

Part of the premier Indian language translation initiative by IIT Madras and AI4Bharat. Contains mined, filtered, and deduplicated sentences tailored for IndicTrans2 fine-tuning.

Usage: Hugging Face: load_dataset('ai4bharat/indic-trans-v2-raw', 'san-eng') | GitHub: github.com/AI4Bharat/IndicTrans2

1.3 AI4Bharat Samantar Indic-English Corpus (Sanskrit Sub-corpus)

Domain: Diverse Web-Mined, Literature, General Knowledge | Volume: 25,000+ parallel sentence pairs

Link: https://indicnlp.ai4bharat.org/samantar/

The largest public parallel corpus for Indic languages. The Sanskrit subset includes sentences extracted from news, open education repositories, and literary texts.

Usage: Direct download via Hugging Face or IndicNLP web portal in TSV / JSONL format.

1.4 OPUS Tatoeba Sanskrit-English Corpus

Domain: Everyday Conversational & Grammatical Sentences | Volume: ~5,000 verified sentence pairs

Link: https://tatoeba.org/en/sentences/show_all_in/san/eng

Community-contributed and linguistic expert-verified Sanskrit sentences paired with English translations. Ideal for idiomatic expressions and colloquial conversational Sanskrit.

Usage: Download pre-packaged TSV from OPUS: opus.nlpl.eu/Tatoeba.php or query the Tatoeba API.

1.5 Meta AI NLLB Flores-200 Evaluation Benchmark

Domain: Multi-Domain News, Science & Cultural Benchmark | Volume: 3,012 gold-standard evaluation sentences (dev & devtest splits)

Link: https://github.com/facebookresearch/flores/tree/main/flores200

Standard multi-domain benchmark used by Meta No Language Left Behind (NLLB) to measure translation quality (BLEU, chrF++, COMET) across 200+ languages including Sanskrit.

Usage: Python: load_dataset('facebook/flores', 'san_Deva-eng_Latn') for standardized test metrics.

1.6 Meta AI WikiMatrix Sanskrit-English

Domain: Wikipedia Parallel Articles & General Knowledge | Volume: 3,500+ parallel sentences mined with LASER multilingual embeddings

Link: https://opus.nlpl.eu/WikiMatrix.php

Parallel sentences mined from Wikipedia using cross-lingual sentence embeddings with cosine similarity > 1.04. High coverage of geographical, historical, and philosophical concepts.

Usage: Download bitext file: WikiMatrix.v1.en-sa.txt.gz from OPUS project page.

1.7 JNU School of Sanskrit and Indic Studies Parallel Corpus

Domain: Panchatantra, Hitopadesha, Gita, News Bulletins | Volume: 50,000+ aligned sentences

Link: http://sanskrit.jnu.ac.in/

Curated by Prof. Girish Nath Jha and SCSS JNU. Features aligned texts across classical fables and contemporary All India Radio Sanskrit news broadcasts.

Usage: Request access or download research packages directly from the JNU computational Sanskrit portal.

1.8 TDIL / MeitY Anuvadaksh Sanskrit Corpus (Govt. of India)

Domain: Administrative, Health, Tourism, and Classical Literature | Volume: 30,000+ domain-tagged parallel sentences
Link: https://tdil-dc.in/
Official Government of India repository developed under the Technology Development for Indian Languages programme. Tagged with parts of speech and domain classification.
Usage: Free academic registration at TDIL Data Centre (tdil-dc.in) to download verified domain bitexts.

2. Epic, Shloka & Philosophical Aligned Bitexts

Verse-by-verse and chapter-by-chapter aligned corpora with word breakdowns

2.1 Mahabharata Critical Edition with K. M. Ganguli Translation

Domain: Epic Narrative & Classical Sanskrit Literature | Volume: 100,000+ shlokas across 18 Parvas with complete English prose
Link: https://www.sacred-texts.com/hin/maha/
Full digitized Sanskrit text of the Mahabharata aligned with Kisari Mohan Ganguli's authoritative English translation. Unlocks vast vocabulary and complex grammatical sandhi contexts.
Usage: GitHub Text Repository: github.com/indic-transliteration/sanskrit-texts or Sacred Texts archives.

2.2 Valmiki Ramayana Aligned Corpus (IIT Kanpur / IITK)

Domain: Epic Poetry, Archaic Sanskrit & Classical Syntax | Volume: 24,000 shlokas across 6 Kandas with word-for-word analysis
Link: https://www.valmikiramayan.net/
Complete text of the Valmiki Ramayana featuring Devanagari verse, Roman transliteration, word-by-word meaning (Padachheda), and comprehensive English translation.
Usage: Scrapeable and downloadable via open repositories and academic mirrors (gitasupersite.iitk.ac.in).

2.3 Gita Supersite Aligned Translation Corpus (IIT Kanpur)

Domain: Vedanta, Philosophy & Didactic Poetry | Volume: 700 shlokas with 7+ English translations and commentaries
Link: https://www.gitasupersite.iitk.ac.in/
Comprehensive philosophical corpus aligning Bhagavad Gita shlokas with translations by Swami Sivananda, Gambhirananda, Purohit Swami, Radhakrishnan, and Ramanuja.
Usage: Open API and public data dumps available from IIT Kanpur Gita Supersite project.

2.4 Principal Upanishads Aligned Corpus (Advaita Sharada)

Domain: Vedic Prose & Verse, Metaphysics | Volume: 108 Upanishads with Shankara Bhashya and English glosses
Link: https://advaitasharada.sringeri.net/
High-fidelity digital preservation of major Upanishads (Isha, Kena, Katha, Prashna, Mundaka, Mandukya, Taittiriya, Chandogya, Brihadaranyaka) with classical and modern English renderings.
Usage: Explore Advaita Sharada digital repository or the Clay Sanskrit Library editions.

2.5 Panchatantra & Hitopadesha Fable Parallel Corpus

Domain: Narrative Prose & Didactic Moral Shlokas | Volume: 250+ fables with aligned English translations (Arthur W. Ryder & Charles Wilkins)
Link: https://www.sanskrit-trikashaivism.com/
Contains continuous narrative prose interspersed with moral couplets, providing essential training data for both running Sanskrit sentences and metric verses.
Usage: Public domain texts available via Project Gutenberg, Internet Archive, and GitHub sanskrit-texts.

2.6 Subhashita Ratnabhandagara Aligned Database

Domain: Subhashitas, Epigrams & Poetic Maxims | Volume: 10,000+ witty verses with English translations and thematic tags
Link: https://github.com/subhashita/
Comprehensive collection of Sanskrit wise sayings, riddles, and ethical verses categorized by topic (friendship, courage, knowledge, dharma) with verified English meanings.
Usage: Clone the Subhashita GitHub repository: git clone https://github.com/subhashita/subhashita.git

3. Bilingual Lexicons & Word-Level Dictionaries (Sanskrit <-> English)

Essential for zero-shot word translations, out-of-vocabulary coverage, and dictionary augmentation

3.1 Cologne Digital Sanskrit Dictionaries (CDSD - University of Cologne)

Domain: Comprehensive Historical & Etymological Lexicons | Volume: 38 major dictionaries (300,000+ headwords with rich definitions)
Link: https://www.sanskrit-lexicon.uni-koeln.de/
The gold standard in Sanskrit lexicography: includes Monier-Williams (1899), Apte Practical Dictionary, Macdonell Practical Dictionary, Cappeller Dictionary, and Shabda-Kalpadruma.
Usage: Direct download in XML, TEI, SQLite, and JSON: sanskrit-lexicon.uni-koeln.de/download/web/webtc/download.html

3.2 Spoken Sanskrit Online Dictionary Database

Domain: Contemporary & Conversational Sanskrit-English Lexicon | Volume: 400,000+ vocabulary entries, inflections, idioms, and compounds
Link: https://spokensanskrit.org/
Extensive user-contributed and verified dictionary covering both classical vocabulary and modern neologisms (science, technology,

politics, daily conversation).
Usage: Query via REST API or download structured vocabulary lists from spokensanskrit.org.

3.3 Sanskrit WordNet (IIT Bombay / CFILT)

Domain: Hierarchical Semantic Lexical Network | Volume: 35,000+ synsets linked to English Princeton WordNet
Link: https://www.cfilt.iitb.ac.in/wordnet/webswn/
Structured lexical database capturing hypernymy, hyponymy, meronymy, antonymy, and semantic roles, linked directly to English synsets for disambiguation.
Usage: Available on request for academic research and NLP projects from CFILT IIT Bombay.

3.4 Heritage du Sanskrit Knowledge Base (Gerard Huet / INRIA)

Domain: Morphological Lexicon, Sandhi Engine & Grammatical Dictionar | Volume: 25,000+ roots and stems with complete declension/conjugation paradi
Link: https://sanskrit.inria.fr/
Exhaustive computational morphology lexicon with French and English glosses, developed over 25+ years by Gerard Huet. Indispensable for Lemmatization and Sandhi splitting.
Usage: Download Zenodo archive or access via INRIA Heritage engine (sanskrit.inria.fr/DICO/index.html).

3.5 Amarakosha Knowledge Engine (University of Hyderabad)

Domain: Ancient Thesaurus with Modern Ontological Tagging | Volume: 10,000+ synsets with English equivalents, gender, and grammatical notes
Link: http://sanskrit.uohyd.ac.in/amara/
The ancient synonymic thesaurus of Amarasimha digitized with structural ontology, grammatical features, and modern English definitions by Prof. Amba Kulkarni.
Usage: Access online or download tagged synset files from the Department of Sanskrit Studies, Univ. of Hyderabad.

4. Grammatical, Morphological & Treebank Resources

Deep syntactic annotations, verbal root paradigms, and dependency parses

4.1 Paninian Dhatupatha Database (Verbal Roots & Conjugations)

Domain: Paninian Grammar & Verbal Semantics | Volume: 2,014 verbal roots with English meanings, Gana, Pada, and models
Link: https://ashtadhyayi.com/dhatu/
Structured database of all Paninian Dhatus (roots) including transitive/intransitive markings, karma/bhava paradigms, and English translation of core verbal actions.
Usage: Download structured JSON / SQLite dumps from Ashtadhyayi.com open API.

4.2 Universal Dependencies Sanskrit Treebanks (sa_ufal & sa_vedic)

Domain: Syntactic Dependency Parsing & Morphological Features | Volume: 6,000+ sentences with CoNLL-U annotations and English lemma glosses
Link: https://universaldependencies.org/treebanks/sa_ufal/index.html
High-quality dependency treebanks (Panchatantra stories in sa_ufal, Rigvedic hymns in sa_vedic). Each word has lemma, UPOS, XPOS, morphological features, and dependency relations.
Usage: Direct GitHub: github.com/UniversalDependencies/UD_Sanskrit-UFAL and UD_Sanskrit-Vedic.

4.3 Ashtadhyayi.com Sutra Knowledge Base

Domain: Formal Paninian Generative Grammar | Volume: 3,983 sutras with Kashika commentary, Padachheda, and English notes
Link: https://ashtadhyayi.com/
The foundational generative grammar of Sanskrit in structured format. Every sutra is annotated with word breakdown, cross-references, rule exceptions, and English explanations.
Usage: Open API access and JSON exports from ashtadhyayi.com/sutra/.

5. Pre-Trained Translation Models for Fine-Tuning & Distillation

Foundation sequence-to-sequence models supporting Sanskrit-English translation

5.1 AI4Bharat IndicTrans2 (1B & 200M Transformer Models)

Domain: State-of-the-Art Indic Machine Translation | Volume: Model Checkpoints: ai4bharat/indictrans2-indic-en-1B and 200M
Link: https://huggingface.co/ai4bharat/indictrans2-indic-en-1B
Trained on 1+ billion pairs across 22 scheduled Indian languages. Achieves top BLEU scores on Sanskrit-to-English translation. Fully compatible with Hugging Face Transformers.
Usage: pip install transformers; model = AutoModelForSeq2SeqLM.from_pretrained('ai4bharat/indictrans2-indic-en-1B', trust_remote_code=True)

5.2 Meta AI NLLB-200 (600M, 1.3B, 3.3B Parameter Checkpoints)

Domain: Massively Multilingual Translation Model | Volume: Supports san_Deva <-> eng_Latn directly with zero-shot generalization
Link: https://huggingface.co/facebook/nllb-200-distilled-600M
Meta's flagship translation model supporting 200 languages. The distilled 600M parameter model runs smoothly on standard GPUs and modern CPUs with sub-second inference.
Usage: Hugging Face: from transformers import AutoTokenizer, AutoModelForSeq2SeqLM; tokenizer.src_lang = 'san_Deva'

5.3 Google mT5 / mBART-50 Many-to-Many Multilingual Seq2Seq

Domain: Multilingual Pre-Trained Sequence-to-Sequence Backbone | Volume: google/mt5-base and facebook/mbart-large-50-many-to-many-mmt
Link: https://huggingface.co/google/mt5-base
Versatile multilingual models pre-trained on Common Crawl mC4. Highly receptive to domain adaptation and fine-tuning on Buddhist-NLP and Itihasa parallel corpora.
Usage: Easily fine-tuned using Hugging Face Seq2SeqTrainer with LoRA / PEFT for rapid training on modest hardware.

6. Practical Fine-Tuning & Ingestion Recipe

How to download, clean, and integrate parallel datasets into this project

Step 1: Download Parallel Bitexts
Use the Hugging Face datasets library to pull buddhist-nlp/skt-en-itihasa or ai4bharat/indic-trans-v2-raw:

```
ds = load_dataset('buddhist-nlp/skt-en-itihasa', split='train')
```

Step 2: Preprocess & Normalize Devanagari
Standardize punctuation: convert ASCII pipes '|' or double pipes '||' to standard Sanskrit single and double dandas.
Apply Unicode NFC normalization to unify decomposed vowel signs, anusvaras, and consonants.

Step 3: Dictionary & In-Memory Hash Lookup
Compile frequent parallel sentence pairs into a JSON hash map (like sanskrit_expanded_corpus.json).
In translate.py, checking an in-memory cache drops translation latency from ~2,500 ms down to < 1 ms!

Step 4: Neural Model Fine-Tuning with LoRA (PEFT)
For translating novel out-of-corpus sentences, fine-tune nllb-200-distilled-600M or indictrans2-indic-en-1B using Parameter-Efficient Fine-Tuning (LoRA / QLoRA) on your target parallel dataset.